

## 11. Covariance, correlation

Recall: Let  $X, Y$  be jointly absolutely continuous random variables. We write  $f(x, y)$  for the joint density and  $f_X(x), f_Y(y)$  for the marginal densities respectively.

Let  $g: \mathbb{R}^2 \mapsto \mathbb{R}$ , then  $E[g(X, Y)] = \iint g(x, y) f(x, y) dx dy$

Prop: Let  $X, Y$  be jointly absolutely continuous, independent random variables. Let  $g: \mathbb{R} \mapsto \mathbb{R}, h: \mathbb{R} \mapsto \mathbb{R}$ , then

$$E[g(X) \cdot h(Y)] = E[g(X)] \cdot E[h(Y)].$$

proof: They are independent, thus  $f(x, y) = f_X(x) f_Y(y)$ .

$$\begin{aligned} E[g(X) h(Y)] &= \iint g(x) h(y) f(x, y) dx dy = \iint g(x) h(y) f_X(x) f_Y(y) dx dy \\ &= \int g(x) f_X(x) dx \int h(y) f_Y(y) dy = E[g(X)] \cdot E[h(Y)] \quad \square \end{aligned}$$

def: Let  $X, Y$  be random variables. We define the covariance of  $X$  and  $Y$  as  $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$

Prop:  $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$

proof:  $\underbrace{E[(X - E[X])(Y - E[Y])]}_{\text{Cov}(X, Y)} \overset{\substack{\uparrow \\ \text{linearity of } E}}{=} E[XY] - E[X]E[Y] - \underbrace{E[X]E[Y]}_{E[X]E[Y]} + \underbrace{E[E[X]E[Y]]}_{E[X]E[Y]}$  □

Cor. 1 If  $X, Y$  are independent, then  $\text{Cov}(X, Y) = 0$ .

proof: Independence  $\Rightarrow E[XY] = E[X]E[Y] \Rightarrow \text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$ . □

Note: It does not hold the other way around! Zero covariance doesn't imply independence! □

Example:  $X = \begin{cases} -1, & \text{prob.} = 1/3 \\ 0, & \text{prob.} = 1/3 \\ 1, & \text{prob.} = 1/3 \end{cases}$

$$Y = X^2$$

$$E[X] = 0 \text{ by definition}$$

$$E[X] = (-1) \cdot 1/3 + 0 \cdot 1/3 + 1 \cdot 1/3 = 0$$

$$E[XY] = E[X^3] = (-1)^3 \cdot 1/3 + 0^3 \cdot 1/3 + 1^3 \cdot 1/3 = 0$$

$$\Rightarrow \text{Cov}(X, Y) = \underbrace{E[XY]}_{=0} - \underbrace{E[X]}_{=0} \underbrace{E[Y]}_{<\infty} = 0, \text{ but } X, Y \text{ are obviously NOT independent.}$$

Prop. The covariance is

1/ positive definite:  $\text{Cov}(X, X) = D^2[X] \geq 0$ ,  $D^2[X] = 0 \Leftrightarrow P(X = E[X]) = 1$

1/ symmetric:  $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

1/ bilinear: For all  $\{a_i, c_i\}_{i \in I} \subseteq \mathbb{R}$ ,  $\{b_j, d_j\}_{j \in J} \subseteq \mathbb{R}$  with  $I, J$  finite:

$$\text{Cov}\left(\sum_{i \in I} (a_i X_i + c_i), \sum_{j \in J} (b_j Y_j + d_j)\right) = \sum_{i \in I} \sum_{j \in J} a_i b_j \text{Cov}(X_i, Y_j)$$

proof. 1/ and 1/ are easy consequences of the definition.

$$\text{1/ } E\left[\left(\sum_{i \in I} (a_i X_i + c_i) - E\left[\sum_{i \in I} (a_i X_i + c_i)\right]\right)\left(\sum_{j \in J} (b_j Y_j + d_j) - E\left[\sum_{j \in J} (b_j Y_j + d_j)\right]\right)\right] \stackrel{\text{linearity of } E}{=}$$

$$= E\left[\left(\sum_{i \in I} a_i X_i - \sum_{i \in I} a_i E[X_i]\right)\left(\sum_{j \in J} b_j Y_j - \sum_{j \in J} b_j E[Y_j]\right)\right] =$$

$$= E\left[\left(\sum_{i \in I} a_i (X_i - E[X_i])\right)\left(\sum_{j \in J} b_j (Y_j - E[Y_j])\right)\right] = E\left[\sum_{i \in I} \sum_{j \in J} a_i b_j (X_i - E[X_i])(Y_j - E[Y_j])\right] =$$

$$\stackrel{\text{linearity of } E}{=} \sum_{i \in I} \sum_{j \in J} a_i b_j E[(X_i - E[X_i])(Y_j - E[Y_j])] = \sum_{i \in I} \sum_{j \in J} a_i b_j \text{Cov}(X_i, Y_j) \quad \square$$

Note. This property implies that covariance is a scalar product over the vector space of random variables (where the constant random variables are identified with the constant 0 rv.)

def. Let  $X_1, X_2, \dots, X_n$  be random variables. We call

$$\underline{\underline{C}} = (\text{Cov}(X_i, X_j))_{i,j=1}^n$$

the covariance matrix of  $X_1, \dots, X_n$ .

Prop. The covariance matrix is always positive semidefinite.

proof. Let  $\underline{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$ , then  $\underline{a}^T \underline{\underline{C}} \underline{a} = \sum_{i,j} a_i \text{Cov}(X_i, X_j) a_j = \text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^n a_j X_j\right)$

$$= D^2\left[\sum_{i=1}^n a_i X_i\right] \geq 0. \quad \square$$

Prop. Let  $\underline{X}$  be a  $d$ -dimensional jointly normally distributed random variables with parameters  $\underline{m} \in \mathbb{R}^d$  and  $\underline{A} \in \mathbb{R}^{d \times d}$  symmetric, positive definite.

Then the covariance matrix of  $\underline{X}$  is  $\underline{\underline{C}} = \underline{A}^{-1}$ .

proof.  $\underline{A}$  is symmetric and positive definite  $\Rightarrow \underline{A} = \underline{P} \underline{D} \underline{P}^T$   
for some  $\underline{P}$  orthonormal and  $\underline{D}$  diagonal matrices.  
The elements on the diagonal of  $\underline{D}$  are all positive.

We've already shown that

$$\underline{X} = \underline{P} \underline{D}^{-1/2} \underline{Y} + \underline{m}, \text{ where } \underline{Y} = (Y_1, \dots, Y_d)^T \text{ and } Y_1, \dots, Y_d \text{ are iid. standard normals.}$$

(independent identically distributed)

$$\begin{aligned}
 \underline{\text{Cov}}(X_i, X_j) &= \text{Cov}\left(\sum_{k=1}^d P_{ik} D_{kk}^{-1/2} Y_k, \sum_{m=1}^d P_{jm} D_{mm}^{-1/2} Y_m\right) \stackrel{\substack{\text{D is diagonal} \\ \text{Cov is bilinear}}}{=} \sum_{k=1}^d \sum_{m=1}^d P_{ik} D_{kk}^{-1/2} D_{mm}^{-1/2} P_{jm} \underbrace{\text{Cov}(Y_k, Y_m)}_{\substack{1, k=m \\ 0, \text{ else}}} = \sum_{k=1}^d P_{ik} D_{kk}^{-1} P_{jk} \underbrace{\text{Cov}(Y_k, Y_k)}_{= D^2[Y_k] = 1} \\
 &= \sum_{k=1}^d P_{ik} D_{kk}^{-1} P_{jk} = \left(\underline{P} \underline{D}^{-1} \underline{P}^T\right)_{ij} = \underline{(A^{-1})}_{ij} \quad \square
 \end{aligned}$$

Prop. Let  $X_1, X_2, \dots, X_n$  be random variables, then

$$D^2\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n D^2[X_i] + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j)$$

proof.

$$\begin{aligned}
 D^2\left[\sum_{i=1}^n X_i\right] &= \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) = \sum_{i,j=1}^n \text{Cov}(X_i, X_j) = \\
 &= \sum_{i=1}^n \underbrace{\text{Cov}(X_i, X_i)}_{= D^2[X_i]} + \sum_{i=1}^n \sum_{j < i} \underbrace{\text{Cov}(X_i, X_j)}_{= \text{Cov}(X_j, X_i)} + \sum_{i=1}^n \sum_{j > i} \text{Cov}(X_i, X_j) = \sum_{i=1}^n D^2[X_i] + 2 \sum_{i=1}^n \sum_{j > i} \text{Cov}(X_i, X_j) \quad \square
 \end{aligned}$$

Cor. If  $X, Y$  are independent, then

$$D^2[X \pm Y] = D^2[X] + D^2[Y]$$

proof.  $D^2[X \pm Y] = D^2[X] + D^2[Y] - 2 \underbrace{\text{Cov}(X, Y)}_{= 0} \quad \square$

Example: (Experimental expected value/mean value of random variables)

Let  $X_1, \dots, X_n$  be independent identically distributed random variables (iid.)

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \text{ we've seen that } E[\bar{X}] = E[X_1].$$

$$\begin{aligned}
 D^2[\bar{X}] &= D^2\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = E\left[\frac{1}{n^2} \left(\sum_{i=1}^n X_i\right)^2\right] - E^2\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n^2} D^2\left[\sum_{i=1}^n X_i\right] \stackrel{\text{indep}}{=} \\
 &= \frac{1}{n^2} \sum_{i=1}^n D^2[X_i] \stackrel{\text{iid}}{=} \frac{1}{n^2} n D^2[X_1] \xrightarrow[n \rightarrow \infty]{\frac{1}{n}} 0.
 \end{aligned}$$

def. Let  $X_1, \dots, X_n$  be iid. random variables. Their experimental variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Prop. If  $D^2[X_1] = \sigma^2$ , then  $E[S^2] = \sigma^2$ .

proof: Assume without loss of generality that  $E[X_i] = 0$ .

$$\begin{aligned} E[(X_i - \bar{X})^2] &= D^2[X_i - \bar{X}] = D^2\left[\frac{n-1}{n}X_i - \frac{1}{n}\sum_{j=1, j \neq i}^n X_j\right] = D^2\left[\frac{n-1}{n}X_i\right] + \sum_{j=1, j \neq i}^n D^2\left[\frac{1}{n}X_j\right] = \\ &= \left(\frac{n-1}{n}\right)^2 D^2[X_1] + \frac{1}{n^2}(n-1) D^2[X_1] = \left(\frac{n-1}{n}\right)^2 \sigma^2 + \frac{n-1}{n^2} \sigma^2 = \frac{n-1}{n} \sigma^2 \end{aligned}$$

$$\Rightarrow E[S^2] = \frac{1}{n-1} \sum_{i=1}^n E[(X_i - \bar{X})^2] = \frac{1}{n-1} \cdot n \cdot \frac{n-1}{n} \cdot \sigma^2 = \underline{\underline{\sigma^2}} \quad \square$$

Example: (Binomial random variable)

Let  $X \sim \text{Bin}(n, p)$ , and  $X_i = \begin{cases} 1, & \text{if the } i\text{th experiment is successful} \\ 0, & \text{if the } i\text{th experiment fails} \end{cases}$

$$\Rightarrow X = \sum_{i=1}^n X_i, \quad \underline{E[X]} = \sum_{i=1}^n \underline{E[X_i]} = \underline{np}$$

$= P(X_i = 1) = p$

$$\begin{aligned} \underline{D^2[X_i]} &= \sum_{i=1}^n \underline{D^2[X_i]} = \underline{np(1-p)} \\ &= E[X_i^2] - E^2[X_i] \quad \swarrow \text{ } X_i \text{ is an indicator} \\ &= E[X_i] - E^2[X_i] = p - p^2 = p(1-p) \end{aligned}$$

Thm: (Cauchy-Schwartz) Let  $X, Y$  be random variables, then

$$|E[XY]| \leq \sqrt{E[X^2]E[Y^2]}$$

proof: Let  $Z_1 = \frac{X}{\sqrt{E[X^2]}} + \frac{Y}{\sqrt{E[Y^2]}}$ , then

$$0 \leq E[Z_1^2] = E\left[\frac{X^2}{E[X^2]} + \frac{Y^2}{E[Y^2]} + 2 \frac{XY}{\sqrt{E[X^2]E[Y^2]}}\right] = \underbrace{\frac{1}{E[X^2]} E[X^2]}_{=1} + \underbrace{\frac{1}{E[Y^2]} E[Y^2]}_{=1} + 2 \frac{E[XY]}{\sqrt{E[X^2]E[Y^2]}}$$

$$\Rightarrow 0 \leq 2 + 2 \frac{E[XY]}{\sqrt{E[X^2]E[Y^2]}} \Rightarrow \underline{\underline{E[XY] \geq -\sqrt{E[X^2]E[Y^2]}}}$$

Doing the same calculations for  $Z_1' = \frac{X}{\sqrt{E[X^2]}} - \frac{Y}{\sqrt{E[Y^2]}}$  yields  $\underline{\underline{E[XY] \leq \sqrt{E[X^2]E[Y^2]}}}$   $\square$

def: Let  $X, Y$  be random variables. We define their correlation as

$$\text{Corr}(X, Y) = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{D[X]D[Y]}.$$

Note: As a consequence of the Cauchy-Schwartz inequality  $\rho(X, Y) \in [-1, 1]$ .

Prop: If  $\rho(X, Y) \in \{-1, 1\}$ , then  $\exists a, b \in \mathbb{R} : Y = aX + b$

proof: Without loss of generality we may assume  $E[X] = E[Y] = 0$

$$\frac{\text{Cov}(X, Y)}{\sqrt{E[X^2]E[Y^2]}} = \frac{E[XY]}{\sqrt{E[X^2]E[Y^2]}} = 1 \Rightarrow 2 - \frac{2E[XY]}{\sqrt{E[X^2]E[Y^2]}} = 0$$

$$E\left[\left(\frac{X}{\sqrt{E[X^2]}} - \frac{Y}{\sqrt{E[Y^2]}}\right)^2\right] = 0 \quad \leftarrow \text{As in the proof of the Cauchy-Schwarz thm. (just backwards)}$$

$$\Leftrightarrow P\left(\left(\frac{X}{\sqrt{E[X^2]}} - \frac{Y}{\sqrt{E[Y^2]}}\right)^2 = 0\right) = 1 \Leftrightarrow P\left(X = \sqrt{\frac{E[X^2]}{E[Y^2]}} Y\right) = 1.$$

$$\text{Similarly, } \rho(X, Y) = -1 \text{ implies } P\left(X = -\sqrt{\frac{E[Y^2]}{E[X^2]}} Y\right) = 1. \quad \square$$

## I.] Conditional density of normal random variables

Let  $X, Y$  be jointly normal random variables with cov. matrix  $\underline{C}$  and expectation  $\underline{m}$ .

$$f(x, y) = \frac{1}{2\pi \sqrt{\det(\underline{C})}} e^{-\frac{1}{2} (x, y) - \underline{m}^T \underline{C}^{-1} (x, y) - \underline{m}}$$

$$\underline{C} = \begin{pmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix}, \text{ where by definition } \sigma_X^2 = D^2[X], \sigma_Y^2 = D^2[Y]$$

$$\rho \sigma_X \sigma_Y = \text{Cov}(X, Y)$$

$$|\underline{C}| = \sigma_X^2 \sigma_Y^2 - \rho^2 \sigma_X^2 \sigma_Y^2 = \sigma_X^2 \sigma_Y^2 (1 - \rho^2)$$

$$\underline{C}^{-1} = \begin{pmatrix} \frac{1}{\sigma_X^2 (1 - \rho^2)} & \frac{\rho}{\sigma_X \sigma_Y (1 - \rho^2)} \\ \frac{\rho}{\sigma_X \sigma_Y (1 - \rho^2)} & \frac{1}{\sigma_Y^2 (1 - \rho^2)} \end{pmatrix}$$

$$X \sim N(m_X, \sigma_X^2), Y \sim N(m_Y, \sigma_Y^2) \text{ where } \underline{m} = (m_X, m_Y)$$

$$\text{Prop: } X|Y=y \sim N\left(m_X + \frac{\sigma_X}{\sigma_Y} \rho (y - m_Y), (1 - \rho^2) \sigma_X^2\right)$$

proof: To make the formulas simpler we write  $x' = x - m_X, y' = y - m_Y$

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} = \frac{\frac{1}{2\pi \sigma_X \sigma_Y \sqrt{1 - \rho^2}} e^{-\frac{1}{2} \left( \frac{x'^2}{\sigma_X^2 (1 - \rho^2)} - 2 \frac{x' y' \rho}{\sigma_X \sigma_Y (1 - \rho^2)} + \frac{y'^2}{\sigma_Y^2 (1 - \rho^2)} \right)}}{\frac{1}{\sqrt{2\pi} \sigma_Y} e^{-\frac{1}{2} \frac{y'^2}{\sigma_Y^2}}} \\ &= \frac{1}{\sqrt{2\pi} \sigma_X \sqrt{1 - \rho^2}} e^{-\frac{1}{2} \left( \frac{x'^2}{\sigma_X^2 (1 - \rho^2)} - 2 \frac{x' y' \rho}{\sigma_X \sigma_Y (1 - \rho^2)} + \frac{y'^2}{\sigma_Y^2 (1 - \rho^2)} - \frac{y'^2}{\sigma_Y^2} \right)} = \frac{1}{\sqrt{2\pi} \sigma_X \sqrt{1 - \rho^2}} e^{-\frac{1}{2} \left( \frac{x'^2}{\sigma_X^2 (1 - \rho^2)} - \frac{y'^2 \rho^2}{\sigma_Y^2 (1 - \rho^2)} \right)} \\ &= \frac{1}{\sqrt{2\pi} \sigma_X \sqrt{1 - \rho^2}} e^{-\frac{1}{2} \frac{\left( x' - \frac{(y - m_Y) \rho \sigma_X}{\sigma_Y} \right)^2}{\sigma_X^2 (1 - \rho^2)}} \quad \square \end{aligned}$$

Cor: If  $X, Y$  are jointly normal with  $\text{Cov}(X, Y) = 0$ , then they are independent.

proof:  $X \sim N(m_X, \sigma_X^2), Y \sim N(m_Y, \sigma_Y^2), \rho = 0 \Rightarrow X|Y=y \sim N(m_X, \sigma_X^2)$  (for some parameters  $m_X, m_Y, \sigma_X, \sigma_Y$ )

$$\Rightarrow f(x, y) = f_{X|Y}(x|y) f_Y(y) = f_X(x) f_Y(y) \Rightarrow \text{they're independent.} \quad \square$$